

Scalability & Future Plan

Date	15 JULY 2024
Team ID	
Project Name	AI-Based Credit Card Approval Prediction System
Maximum Marks	1Mark

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Model trained on a limited historical dataset.	May reduce prediction accuracy for new or diverse applicant profiles.	High
2	The system predicts only approval or rejection without explaining the decision.	Users and analysts cannot understand the reasoning behind predictions.	High
3	Supports single-user prediction at a time.	Supports single-user prediction at a time.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for limited concurrent users.	Deploy using cloud infrastructure with load balancing and auto-scaling.
2	Data Storage	Local dataset and model files.	Integrate cloud databases such as IBM Db2, MongoDB Atlas, or PostgreSQL for scalable data management.
3	Performance	Predictions are generated using a locally stored model.	Optimize prediction pipeline, cache frequently used resources, and use optimized model-serving APIs.
4	Security	Basic Flask application security.	Implement user authentication, HTTPS, API security, encryption, and role-based access control.

Future Road map:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate Explainable AI (SHAP/LIME) for prediction explanations.	2 Months	Improves transparency and user trust.
Phase 2	Develop REST API for banking applications and mobile integration.	3 Months	Enables integration with external banking systems.
Phase 3	Retrain the model periodically using updated banking datasets.	4 Months	Improves prediction accuracy and adaptability.
Phase 4	Deploy on a scalable cloud platform with real-time analytics dashboard and fraud detection.	6 Months	Supports enterprise-level banking applications and large-scale deployment.